2019 EXAMINATIONS

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.306 Internet Applications Engineering

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a A web browser must fetch any linked content that is required to render a page before it can proceed. Describe what this linked content affects and identify two different resources that are considered to be *render blocking*. **[3 marks]**

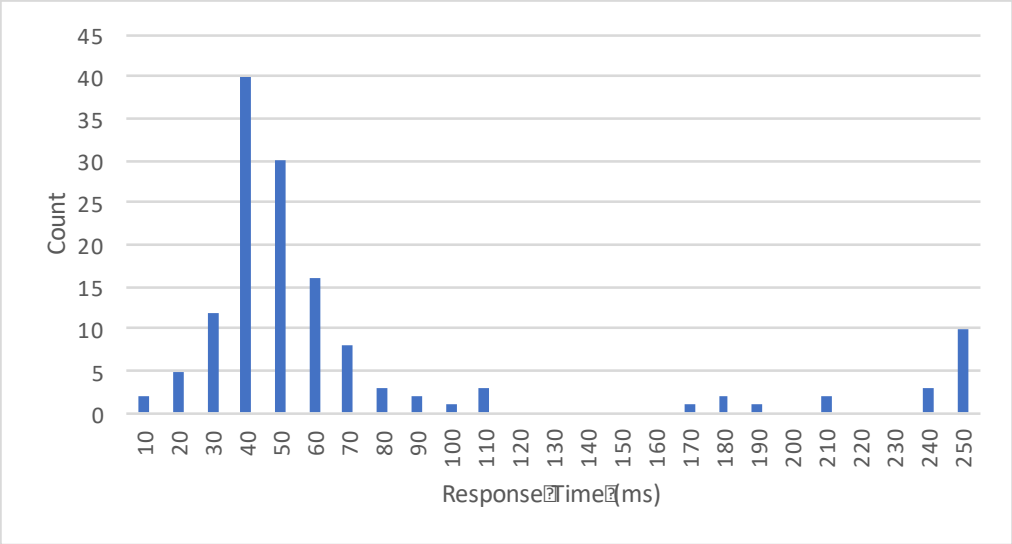
1.b Consider the following HTML code. Using your knowledge of web performance optimisation, apply two techniques to this code to reduce the page load time. In each case, state the approach used, why it improves performance and the line number where it is to be applied.

```
1     <!DOCTYPE html>
2     <html>
3     <head>
4           <script src="js/jquery-3.3.1.full.js"></script>
5     </head>
6     <body>
7           <link rel="stylesheet" type="text/css" href="mystyle.css">
8           <h1>My Web Page</h1>
9           <p>A paragraph.</p>
10    </body>
11    </html>
```

[6 marks]

1.c Performance metrics can be used to diagnose and optimise web sites.

i. Consider the histogram below, which shows the response time for a web server. What can be inferred from this figure? (3 marks)



ii. Basing your findings on a *single* measurement from a *single* location carries significant risks. Describe these risks and explain what can be done to mitigate them. (3 marks)

Question 1 continues on next page…

Question 1 continued.

iii. In addition to response time, other metrics can be used to monitor websites. Describe two of these, including details of how they are calculated. (3 marks)

[9 marks]

1.d Caching is an important process in ensuring that performance requirements are met, especially at scale.

i. Explain the principles of caching, including two reasons why it is effective and the potential locations where caching can be used. (4 marks)

ii. Caching is not appropriate for all types of content. Describe two examples of content that *can* be cached, and one that *cannot*, being clear why they belong to each group. (3 marks)

[7 marks]

Total 25 marks

Question 2

2.a Web Analytics are often used by companies to improve a customer's experience of their website.

i. Describe two of the three *main* metrics that are used in the process. (4 marks)

ii. Describe and compare *Bounce Rate, Exit Rate and Click Through Rate* metrics, being clear on their similarities and differences in each case. (3 marks)

[7 marks]

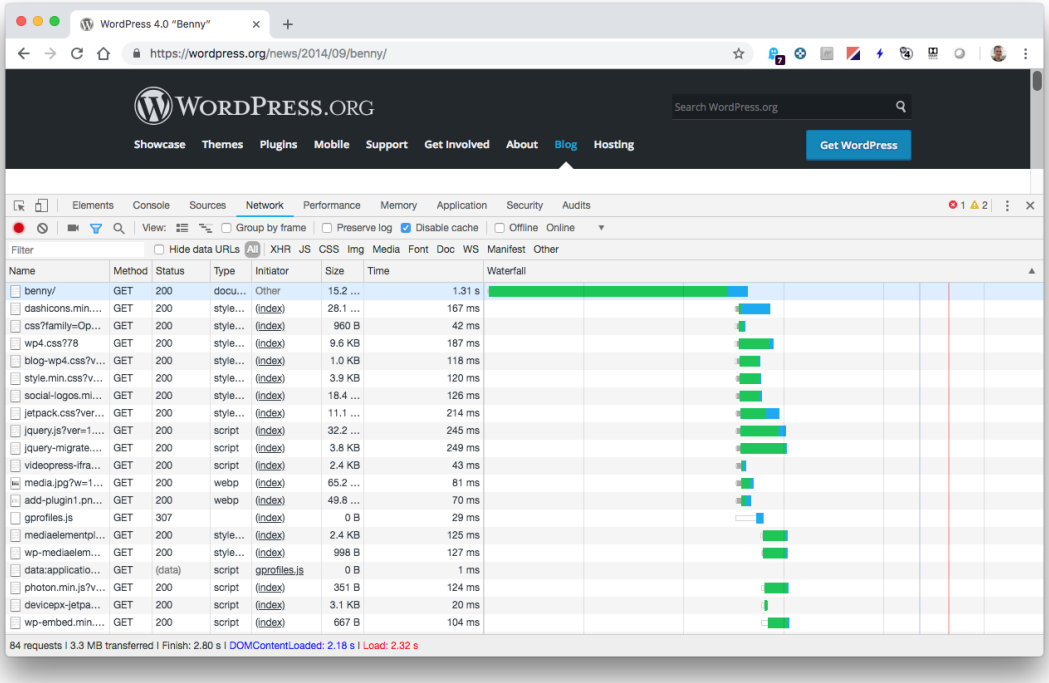
2.b A developer wants to experiment with changing the *font size*, *colour* and *layout* on a single web page. Applying your knowledge of A/B testing, outline the steps that you would take to ensure that these changes have the maximum possible benefit. Use an example to aid your explanation.

[6 marks]

2.c Compare the role of a DNS server to that of a web server. Explain how the two are used to deliver a requested web page.

[3 marks]

2.d Waterfall diagrams are used to help debug website performance. Consider the waterfall diagram below, taken from Chrome Developer Tools.



i. Describe what each row represents. (1 mark)

Question 2 continues on next page…

Question 2 continued.

ii. Notice that the first line appears different, with an increased *Time to First Byte*. Explain what may cause this, and describe how the process of *scaling* could be used to remedy this. Be clear on the different methods, including the positive and negative aspects of each. (6 marks)

iii. Explain the significance of the *DOMContentLoaded* value in terms of user experience and suggest a technique that can be used to minimise its impact. (2 marks)

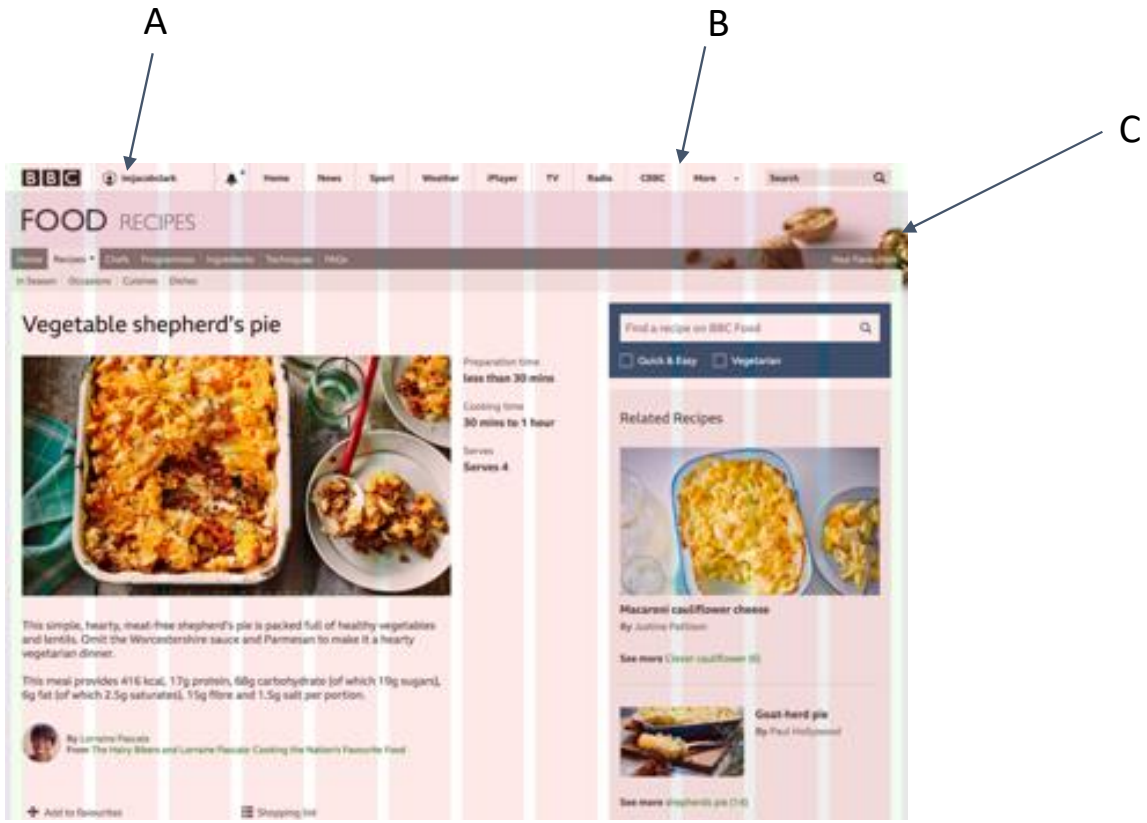
[9 marks]

Total 25 marks

Question 3

3.a Fluid grids allow content, text and images to freely move around a page in response to t he devices screen size.

i. The following diagram shows the three elements that make up a fluid grid (labelled A B and C). Identify each label in the figure below and explain its purpose. (3 marks)



ii. Briefly describe three front-end techniques that exist to create grids. What level of support is available in web browsers for each technique? (3 marks)

iii. Atomic design is a new methodology to help web developers and designers to create websites. Explain the idea behind atomic design, identifying the four parts of the methodology. How does atomic design benefit organisations? (7 marks)

[13 marks]

Question 3 continues on next page…

Question 3 continued.

3.b A web developer for a new local restaurant has produced a mock—up of their site using a responsive design approach. The HTML code they have written is as follows:

```
<html>
<head>
<style>
* {
  box-sizing: border-box;
}

[class*="col-"] {
  float: left;
  width: 100%;
}

[class] {
  border:1px solid black;
}

@media only screen and (min-width: 600px) {
    .col-t-1 {width: 33.33%;}
    .col-t-2 {width: 66.66%;}
    .col-t-3 {width: 100%;}
}
@media only screen and (min-width: 768px) {
    .col-d-1 {width: 33.33%;}
    .col-d-2 {width: 66.66%;}
    .col-d-3 {width: 100%;}
}
</style>
</head>
<body>

  <div class="col-d-3 col-t-3">
    <h1>Cafe 21</h1>
  </div>

  <div class="col-d-1 col-t-1">
    <h2>Starters</h2>
  </div>

  <div class="col-d-1 col-t-2">
    <h2>Mains</h2>
  </div>

  <div class="col-d-1 col-t-3">
    <h2>Desserts</h2>
  </div>

</body>
</html>
```

Question 3 continues on next page…

Question 3 continued.

Study the HTML code above and draw the basic layout of the site as it would appear on devices with the following resolution widths:

i. 400px
ii. 600px
iii. 768px

[6 marks]

3.c A web company that you work for is looking to move their systems over to a cloud hosting provider, such as Amazon Web Services. Discuss the trade-offs that the company will need to consider when using such a service in respect to *Content*, *Speed* and *Infrastructure Cost*.

[6 marks]

Total 25 marks

Question 4

4.a A retailer is looking to improve the security of their website that offers products for customers to purchase online. You have been commissioned to help them with this task. Based on your knowledge of web security, identify the *five* most important threats you believe they should consider, together with a detailed explanation as to how they should address each of them.

[10 marks]

4.b. An important part of engineering software with quality is to *engineer for failure*. Consider the deployment of a web-based system that will be hosted within a cloud environment. Describe *five* approaches that you believe represent good practice when engineering such a system.

[5 marks]

4.c The BBC, on average, logs 760GB and monitors 1 million KPIs each day.

i. In terms of their *systems* infrastructure, *what* different aspects might they monitor, and *why*? (3 marks)

ii. Compare and contrast metrics vs. logs. (4 marks)

iii. An organisation is looking to make increased use of logs within their system. Based on your understanding of how the BBC currently logs information at scale, what *three* recommendations would you give to the organisation in respect to log management? (3 marks)

[10 marks]

Total 25 marks

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